

Simon Chen

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EDUCATION

University of Alberta

Bachelor of Science in Computer Science, Specialization

Sep. 2018 – June 2024

Edmonton, AB

AI & TECHNICAL SKILLS

AI Engineering: Code Synthesis, Heuristic Search, Pathfinding, MCMC, Prompt Engineering, AI-Assisted Development

AI Tools: OpenAI Codex, ChatGPT, Claude, Gemini, Agentic Coding Workflows, AI Code Review, Test Planning

Languages: TypeScript, JavaScript, Python, C++, C#, Java, SQL, HTML, CSS, C

Frontend: Angular, React, Next.js, React Native, Expo, Tailwind CSS, WXML, WXSS

Backend & Cloud: Node.js, AWS Lambda, API Gateway, Serverless Framework, REST APIs, Docker, Linux, CI/CD

Data & Storage: PostgreSQL, MySQL, MongoDB, Redis, DynamoDB, Amazon S3, Data Validation, Matplotlib

EXPERIENCE

Kitchen Worker

Kent Institution

Jul 2026 – Current

Agassiz, BC

- Perform dishwashing, food preparation, sanitation, and inventory duties in a correctional facility kitchen while following strict security, tool-control, and safety procedures

Solar Systems Technician

HSE Solar

Jan 2026 – June 2026

Vancouver, BC

- Configured APSystems firmware devices, monitoring apps, network connectivity, and telemetry diagnostics for solar panel system health tracking
- Installed solar panel systems by assembling railings, mounts, microinverters, panels, and wiring while coordinating safe rooftop workflows

Freelance Creative Technologist

Programming / Digital Art / Video Editing

Dec 2024 – Dec 2025

Vancouver, BC (Remote)

- Managed Google Ads and Analytics, SEO and keyword research, HTML-based website/content updates, and performance analysis to improve digital campaigns within defined advertising budgets
- Created and published video and social/e-commerce content across YouTube, X/Twitter, Instagram, and TikTok, coordinating digital assets and workflows from concept through delivery

Full-Stack Software Engineer Intern

Titan Logix Corp.

Jan 2022 – Sep 2022

Edmonton, AB

- Built and maintained a production Angular and TypeScript dashboard for 1,500+ active users, improving page load performance by 35% through component optimization and API caching
- Developed AWS serverless services with Lambda, API Gateway, DynamoDB, S3, and Serverless Framework CI/CD workflows, supporting 2M+ monthly API requests with 99.9% uptime
- Designed REST APIs and telemetry data pipelines connecting 10,000+ IoT devices for real-time truck sensor monitoring, diagnostics, anomaly review, and fleet data visualization
- Processed high-frequency sensor data for pressure, level, and temperature readings, improving liquid level accuracy by 28% through data quality checks

Artificial Intelligence Researcher

University of Alberta

May 2021 – Sep 2021

Edmonton, AB

- Researched code synthesis for generating stronger heuristic functions before LLM coding assistants became mainstream, improving combinatorial search efficiency by 22–35%
- Built Python evaluation pipelines for Dijkstra, A*, IDA*, and Metropolis-Hastings across 1,000+ sliding-tile puzzle instances to validate synthesized heuristic quality
- Scaled multiprocessing experiments on Compute Canada HPC clusters, applying early program-synthesis ideas now central to agentic AI coding workflows

Frontend Software Engineer Intern

YOYA Network Technology Co. Ltd

May 2020 – Sep 2020

Xiamen, Fujian, China

- Developed a WeChat Mini Program serving 50K+ monthly active users, shipping 20+ mobile features with WXML, WXSS, JavaScript, and API integrations
- Built 30+ reusable UI components through code review, testing, and debugging, cutting delivery time by 40% while improving interface consistency
- Integrated WeChat APIs for authentication, payment, QR scanning, and location services, contributing to a 25% checkout conversion lift

PROJECTS

C++ 2D Game Engine

C++, SFML, Dear ImGui, Entity Component System

github.com/schen32

- Built a custom 2D game engine in modern C++ with SFML rendering/input, Dear ImGui debugging tools, and modular gameplay systems
- Architected a data-oriented ECS supporting 10,000+ active entities per scene with reusable rendering, movement, collision, and state-update systems